CS636 - Formal Methods (Quiz No. 2) **Quiz Start Time: 04:28 PM, 13 May 2024**

Question # 4 of 10 (Start time: 04:32:30 PM, 13 May 2024) **Total Marks:** 1

Modeling phase formalize the property to be checked using the property .....

Select the correct option

Reload Math Equations

- ( ) Binary Language
- ( ) Specification language
- (•) Description language
- ( ) General language

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 2) **Quiz Start Time: 04:28 PM, 13 May 2024**

Question # 5 of 10 (Start time: 04:33:12 PM, 13 May 2024) **Total Marks:** 1

Which of the following is NOT a phase of model checking process

Select the correct option

Reload Math Equations

- ○ Running phase
- ○ Modeling phase
- ◉ Integrated phase
- ○ Analysis phase

CS636 - Formal Methods (Quiz No. 2) **Quiz Start Time: 04:28 PM, 13 May 2024**

Question # 6 of 10 (Start time: 04:34:01 PM, 13 May 2024) **Total Marks:** 1

Generally speaking, a consistency constraint C is a predicate defined on models such that _____ means that the model is consistent with respect to the constraint.

Select the correct option

Reload Math Equations

- M(C) = false
- C(M) = false
- C(M) = true
- M(C) = true

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 2) **Quiz Start Time: 04:28 PM, 13 May 2024**

Question # 7 of 10 (Start time: 04:35:20 PM, 13 May 2024) **Total Marks:** 1

Which one is the correct representation of effect function?

Select the correct option

Reload Math Equations

○	Effect : Act - Eval(Var) → Eval(Var)
✗	Effect : Act + Eval(Var) → Eval(Var)
○	Effect : Act / Eval(Var) → Eval(Var)
👍	Effect : Act × Eval(Var) → Eval(Var)

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 2) **Quiz Start Time: 04:28 PM, 13 May 2024**

Question # 8 of 10 (Start time: 04:36:17 PM, 13 May 2024) **Total Marks:** 1

A model or a diagram is well-formed if its structure satisfies the _____ rules.

Select the correct option

Reload Math Equations

- [ ] study
- [ ] system
- [x] syntactic
- [ ] development

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 04:37:17 PM, 13 May 2024) **Total Marks:** 1

g_0 represents:

Select the correct option

Reload Math Equations

- set of actions
- initial location
- initial condition
- effect function

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 2) **Quiz Start Time: 04:28 PM, 13 May 2024**

Question # 10 of 10 (Start time: 04:37:59 PM, 13 May 2024) **Total Marks:** 1

Model checking is computationally very .....

Select the correct option

Reload Math Equations

- ○ Practical
- ○ Inexpensive
- ● Hard
- ○ Expensive

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 3)

Quiz Start Time: 12:08 PM, 20 May 2024

Question # 1 of 10 (Start time: 12:08:23 PM, 20 May 2024)

Total Marks: 1

p → q is logically equivalent to _____

Select the correct option

Reload Math Equations

- ¬p ∧ q
- p ∨ ¬q
- ¬p ∨ q
- ¬p ∨ ¬q

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 3) **Quiz Start Time: 12:08 PM, 20 May 2024**

Question # 2 of 10 (Start time: 12:09:59 PM, 20 May 2024) **Total Marks:** 1

A compound proposition that is always TRUE and contain only T in their evaluation is called....

Select the correct option

Reload Math Equations

○	Proposition
○	Contingency
◉	Tautologies
○	Contradictions

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 3) **Quiz Start Time: 12:08 PM, 20 May 2024**

Question # 3 of 10 (Start time: 12:10:44 PM, 20 May 2024) **Total Marks:** 1

"if you work hard you will succeed " – is an example for _____ proposition.

Select the correct option

Reload Math Equations

- ○ Emotive
- ○ Positive
- ○ Negative
- ◉ Conditional

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 3) **Quiz Start Time: 12:08 PM, 20 May 2024**

Question # 4 of 10 (Start time: 12:11:30 PM, 20 May 2024) **Total Marks:** 1

Proposition which is affirmed on the basis of premises is called .....

Select the correct option

Reload Math Equations

- ○ Idea
- ○ Concept
- ◉ Conclusion
- ○ Suggestion

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 3) **Quiz Start Time: 12:08 PM, 20 May 2024**

Question # 5 of 10 (Start time: 12:12:37 PM, 20 May 2024) **Total Marks:** 1

Let P: This is a great restaurant, Q: You should not come back here. Then 'This is a great restaurant and you should come back here.' is best represented by?

Select the correct option

Reload Math Equations

- ○ P ∧ Q
- ○ ~P V ~Q
- ○ P V Q
- ◉ P ∧ ~Q

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 3) **Quiz Start Time: 12:08 PM, 20 May 2024**

Question # 6 of 10 (Start time: 12:13:56 PM, 20 May 2024) **Total Marks:** 1

A proposition is said to be ..... if it cannot be broken down into simpler propositions, that is, if it is not composite.

Select the correct option

Reload Math Equations

- (●) Primitive
- ( ) Command
- ( ) Logical
- ( ) Conditional

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 3) **Quiz Start Time: 12:08 PM, 20 May 2024**

Question # 7 of 10 (Start time: 12:15:02 PM, 20 May 2024) **Total Marks:** 1

The universal quantifier denoted by ---------

Select the correct option

Reload Math Equations

- (●) ∀
- ( ) ~
- ( ) ∧
- ( ) ∈

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 3) **Quiz Start Time: 12:08 PM, 20 May 2024**

Question # 8 of 10 (Start time: 12:16:19 PM, 20 May 2024) **Total Marks:** 1

The truth table contains only T in the ____ column in tautology?

Select the correct option

Reload Math Equations

- ( ) Second
- ( ) Initial
- ( ) Middle
- (•) Final

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 3) **Quiz Start Time: 12:08 PM, 20 May 2024**

Question # 9 of 10 (Start time: 12:17:52 PM, 20 May 2024) **Total Marks:** 1

Which of the following is NOT an example of a proposition?

Select the correct option

Reload Math Equations

- ○ 2+6=9
- ○ Ice floats on the water
- ○ 2+2=4
- ◉ Where are you going?

Click to Save Answer & Move to Next Question

CS636 - Formal Methods (Quiz No. 3) **Quiz Start Time: 12:08 PM, 20 May 2024**

Question # 10 of 10 (Start time: 12:19:02 PM, 20 May 2024) **Total Marks:** 1

Conjunction is represented by the English word .....

Select the correct option

Reload Math Equations

- ○ NOT
- ○ OR
- ○ NOR
- ◉ AND

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 12:47:13 PM, 13 May 2024)

Which of the following is NOT a verification technique.

Select the correct option

○	Testing
◉	Integration
○	Model Checking
○	Simulation

Question # 3 of 10 (Start time: 04:41:27 PM, 22 May 2024)

A compound proposition that is always TRUE and contain only T in their evaluation is called....

Select the correct option

◉	Tautologies
○	Contradictions
○	Contingency
○	Proposition

Question # 9 of 10 (Start time: 12:49:31 PM, 13 May 2024)

Act represents:

Select the correct option

●	set of actions
○	set of locations
○	initial condition
○	effect function

Question # 2 of 10 (Start time: 12:41:18 PM, 13 May 2024)

Loc_0 represents:

Select the correct option

○	effect function
○	intial condition
○	set of actions
◉	set of initial locations

Question # 10 of 10 (Start time: 04:46:48 PM, 22 May 2024)

An argument is an assertion that a given set of propositions P1, P2,...,Pn, called .....

Select the correct option

○	Conclusion
●	Premises
○	Contradictions
○	Solution

Question # 8 of 10 (Start time: 04:45:02 PM, 22 May 2024)

In the system of logic, a proposition could have the value TRUE, FALSE or .....

Select the correct option

○	Conditional
○	Fallacy
○	Negative
●	Undefined

Question # 9 of 10 (Start time: 04:46:33 PM, 22 May 2024)

Proposition which is affirmed on the basis of premises is called .....

Select the correct option

◉	Conclusion
○	Concept
○	Idea
○	Suggestion

Question # 4 of 10 (Start time: 04:41:55 PM, 22 May 2024)

The truth table contains only T in the _____ column in tautology?

Select the correct option

○	Middle
○	Second
○	Initial
●	Final

CS636 - Formal Methods (Quiz No. 1)

Question # 3 of 10 (Start time: 10:41:55 PM, 30 April 2024)

Correctness of entire software depends on _____

Select the correct option

- ○ Maintenance
- ○ Deployment
- ○ Testing 
- ○ Specification

CS636 - Formal Methods (Quiz No. 1)

Question # 6 of 10 (Start time: 10:45:36 PM, 30 April 2024)

Formal specifications can help considerably in generating suitable _____

Select the correct option

- ○ mathematical notations
- ○ use cases
- ○ formulae
- ○ test cases

Question # 7 of 10 (Start time: 10:46:51 PM, 30 April 2024) T

A _____ model that represents the intended behavior of a program can be used as a formal specification

Select the correct option

Reload Math

- mathematical
- data
- conceptual
- computational

CS636 - Formal Methods (Quiz No. 1)

Question # 8 of 10 (Start time: 10:48:14 PM, 30 April 2024)

The minimum number of states required to recognize an octal number divisible by 3 are/is

Select the correct option

○	1
○	2
○	3
○	5

CS636 - Formal Methods (Quiz No. 1)

Question # 9 of 10 (Start time: 10:49:04 PM, 30 April 2024)

How many elements in the set for the Language $L=\{x\in(\Sigma r)^* | \text{length if x is at most 2}\}$ and $\Sigma=\{0,1\}$ is___

Select the correct option

○	8
○	6
○	5 ✓
○	7

Question # 5 of 10 (Start time: 10:44:43 PM, 30 April 2024)

Formal specifications allow for feedback on system specifications at _____ development stages.

Select the correct option

Rel

- middle
- later
- early
- early and later

CS636 - Formal Methods (Quiz No. 1) Quiz Start Time: 10:39

Question # 4 of 10 (Start time: 10:42:43 PM, 30 April 2024)

_____ constitute a branch of software engineering that incorporates the use of mathematics for software

Select the correct option

Reload

- Quality assurance
- Requirement engineering
- Formal methods
- Project management

Question # 1 of 10 (Start time: 10:39:39 PM, 30 April 2024)

The language that can be expressed by any regular expression is called a _____

Select the correct option

- Prime Language
- Binary language
- Regular language
- Factorial